YVETTE D. HASTINGS

DATA SCIENTIST

Salt Lake City, UT | Phone: +1(801)520-0991 | Email | Portfolio | GitHub | LinkedIn

SUMMARY

Data Scientist with 5+ years of graduate research experience applying statistical modeling, machine learning, and geospatial analysis to housing, environmental, and software-quality datasets. Proficient in Python, R, and SQL, with experience in regression, outlier analysis, feature engineering, model validation, and analysis of field-collected data. Author of six first-authored peer-reviewed publications, with experience translating complex statistical findings for interdisciplinary researchers and nontechnical audiences.

EDUCATION

Master of Science in Computer Science | *Montana State University* (GPA: 3.81) January 2023 – August 2026

Master of Science in Geography | *University of Utah* (GPA: 3.935) August 2020 – August 2022

TECHNICAL SKILLS

- **Statistical & Data Science Methods**: Statistical Modeling, Experimental Design, Survey Design, Hypothesis Testing, Machine Learning, Data Mining, Exploratory Data Analysis, NLP, Time Series Analysis, Model Evaluation
- **Data Analysis Tools**: R, Python, SQL, Excel, Qualtrics
- **Machine Learning & Analytics**: scikit-learn, XGBoost, pandas, NumPy, Plotly, Matplotlib, dplyr, tidyverse, mlr3, ggplot2
- **Reporting Tools**: Overleaf, MS Office Suite, Adobe Creative Cloud, Markdown
- **Version Control**: Git, GitHub, GitLab

SELECTED PROJECTS

Gallatin County Housing Price Predictions **September 2025 – October 2025**

- Developed and validated predictive models using more than 17,000 residential-property records, including regression, regularization, clustering, random forest, and other supervised and unsupervised methods.
- Engineered features through data cleaning, missing-data treatment, outlier analysis, categorical encoding, and standardization.
- Compared models using five-fold cross-validation, R^2, and RMSE and evaluated feature importance to identify the property characteristics most influential to housing-value predictions.
- Achieved $R^2 = 0.608$ using Lasso and ordinary least-squares regression models.

Mining User Forums to Evaluate Software Quality-in-Use **February 2024 – December 2025**

- Web-scraped 3,941 user forum threads across four forums using R and rvest, then applied NLP and text-mining methods to identify user-centric software quality concerns.
- Developed text-mining lexicons and applied mixed-effects statistical models to identify significant user-reported software quality concerns defined by ISO/IEC 25019.
- Identified software usability as the dominant concern across 68% of analyzed user threads.
- First-author peer-reviewed publication at IEEE Southeast Con 2026.

Public Park Accessibility Analysis **November 2020 – December 2020**

- Integrated U.S. Census demographic and household-income data with Utah housing data using SQL.
- Conducted geospatial network and service-area analyses in ArcGIS Pro to evaluate socioeconomic accessibility among Salt Lake County households.
- Identified geographic and socioeconomic disparities and developed an interactive application to communicate findings to nontechnical audiences.

WORK EXPERIENCE

Montana State University | *Graduate Research Assistant* **January 2023 – May 2026**

- Applied statistical analysis and machine learning with R and Python to prepare and analyze environmental and software engineering datasets, identify significant predictors, and conduct model evaluation against research hypotheses.
- Collaborated with interdisciplinary researchers to design analyses, interpret statistical results, and communicate research findings through five peer-reviewed publications and presentations.

University of Utah | *Graduate Research & Teaching Assistant* **August 2020 – December 2022**

- Performed statistical analyses in R to evaluate environmental research hypotheses and interpret relationships among urban water and ecological variables, contributing to a first-author peer-reviewed publication.
- Analyzed longitudinal, field-collected environmental data using regression, mixed-effects modeling, and piecewise structural equation modeling to evaluate relationships among soil conditions, microbial diversity, biomass, and enzyme activity.
- Designed and developed an environmental monitoring dashboard using R Shiny to improve citizen science and research collaboration.

American West Analytical Labs | *Inorganic Chemistry Assistant Supervisor* **July 2016 – July 2020**

- Trained 30+ employees on EPA and laboratory methods to enhance performance and efficiency.
- Researched and applied EPA methods to create and update laboratory SOPs.
- Reviewed analytical data and client reports for accuracy and compliance with EPA and laboratory standards.
- Streamlined workflow processes to increase overall laboratory efficiency.